



Beyond Blood Sugar: Exploring the Contribution of Metabolic Syndrome to Peripheral Neuropathy

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Diabetic peripheral neuropathy (DPN) is the most common complication of patients with diabetes, occurring in up to 40% of patients with type 2 diabetes mellitus (T2DM) (1,2). Patients with DPN can develop pain, numbness, balance issues, and weakness, leading to significant morbidity and dramatically impacting their quality of life. Despite its high prevalence and impact on patients, there have been no approved medications for the treatment of DPN. In fact, even tightly controlling blood glucose levels in T2DM does not appear to prevent worsening or development of DPN (1,2). Therefore, identification of at-risk individuals with early intervention is currently the best option for avoiding DPN and other complications from T2DM. There are several recognized prediabetes states, all based on blood glucose measurements. The American Diabetes Association defines prediabetes as having impaired fasting glucose, impaired glucose tolerance, or moderate elevation in hemoglobin A_{1c}, which is a measure of average blood glucose over 3 months (3). Though not categorized as prediabetes, metabolic syndrome (MetS) is increasingly recognized as both a precursor to T2DM and an important contributor to the complications of the disease (4). MetS is characterized by obesity, hyperlipidemia, hypertension, and hyperglycemia and, similar to diabetes, is associated with a high prevalence of peripheral neuropathy (PN) (5,6). Through better understanding of the etiology and potential treatments for PN secondary to MetS, researchers hope to have an impact on this condition, as well as DPN. In the original research article in this issue of *Diabetes* entitled “Ketogenic Diet and Exercise Improve Peripheral Neuropathy in a Mouse Model of Metabolic Syndrome,” Eid et al. (7) investigate whether, and by what mechanism, dietary and behavioral changes can alter the metabolic profile and delay or reverse PN in a mouse model for MetS.

MetS is extremely common, with a worldwide prevalence of almost 25% (8). MetS is a major risk factor for

development of both T2DM and PN, with MetS increasing the risk of developing PN by up to 40% (5). Features of MetS also correlate with severity of PN and presence of painful neuropathy (9). Given the functional impact of PN, development of new treatments or lifestyle interventions for MetS neuropathy is critical. Two interventions that have been shown to improve MetS and subsequent PN are exercise (10) and ketogenic diet (11–13). A ketogenic diet, which involves high intake of fats and minimal to no ingestion of carbohydrates, induces a shift from metabolizing glucose to fatty acids and the production of ketones, specifically β -hydroxybutyrate and acetoacetate. A ketogenic diet has been used to treat several neurologic conditions—most prominently intractable epilepsy (14). With respect to PN, the ketogenic diet has not been widely used clinically but has shown promise in improving MetS in patients (13) and PN in mouse models (12). The mechanism by which these interventions impact MetS or PN remains unknown.

In the study by Eid et al., the investigators use a common mouse model of MetS to investigate whether a ketogenic diet contributes to development of MetS and PN, compared with high-fat diet (HFD) and standard diet. They subsequently evaluate whether a ketogenic diet, exercise, or a combination of the two interventions can promote recovery from MetS and PN induced by HFD. Lastly, they investigate sensory neuron cultures and bulk RNA sequencing (RNAseq) from sciatic nerves and muscles to ascertain the mechanisms underlying the development of HFD-induced PN and the impact of the intervention groups. There are several important findings from these experiments. First, lard-based HFD led to MetS in mice, measured as increased fat body mass, decreased lean body mass, and fatty liver, while ketogenic diet, using a cocoa butter fatty diet, did not induce any of these metabolic changes. HFD, but not ketogenic diet, also led to PN, measured by nerve electrophysiology and intraepidermal nerve fiber density. Second, HFD-induced MetS and subsequent

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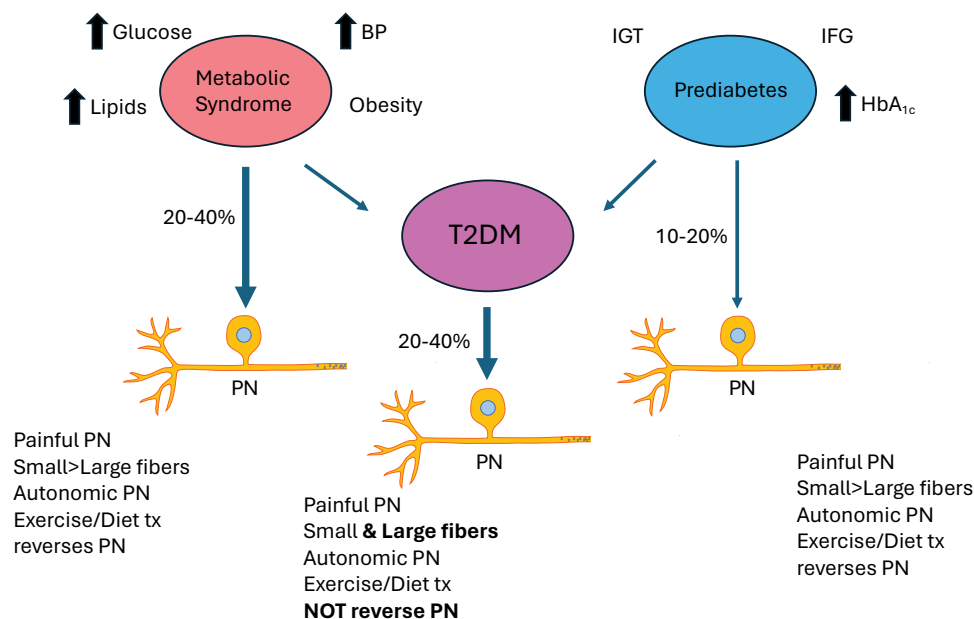


Figure 1—Two medical conditions are recognized as precursors to T2DM: MetS and prediabetes. MetS is defined as having elevated lipids, glucose, blood pressure (BP), body weight, and blood glucose. Patients with MetS have increased risk of developing T2DM and several complications, including PN. Prediabetes is defined by elevated blood glucose: impaired glucose tolerance (IGT), impaired fasting glucose (IFG), or overall elevated blood glucose (measured as hemoglobin A_{1c} [HbA_{1c}]). Patients with prediabetes also have an increased risk of developing T2DM and PN. Many of the clinical characteristics of PN between these three conditions are identical, though importantly, the neuropathy caused by prediabetes and MetS appears to be reversible, while T2DM neuropathy is generally irreversible and has a greater impact on large-fiber peripheral nerves. tx, treatment.

PN were treatable by switching mice to standard diet, a ketogenic diet, exercise, or combined ketogenic diet and exercise. The combined intervention was the most effective and the only one to improve all measured end points of MetS and PN. Third, treatment of neuronal cultures with palmitate, which is a saturated fatty acid prominent in HFD, led paradoxically to impaired ATP production, supporting that MetS causes mitochondrial toxicity, as published previously (15). This impaired cellular energetics could be prevented by pretreating cells with the ketone β -hydroxybutyrate. Finally, RNAseq experiments were completed to characterize gene expression changes produced by HFD, along with the impact of each of the interventions. Though exploratory by design, these studies showed that the differential gene expression in the HFD group, which involved upregulation of immune response, cytoskeletal dynamics, and extracellular matrix interactions, along with downregulation of fatty acid metabolism pathways, was reversed by the interventions. In these experiments specific genetic signatures were also found for each of the interventions, that could be explored in future studies.

There were several important novel conclusions from the study by Eid et al. that are important for this field of study. First, alterations in both metabolism and peripheral nerves from an HFD are reversible. Second, the study provides further evidence for the severity of PN from MetS and the similarities to DPN. Finally, potential genetic pathways involved in MetS PN were identified,

along with genetic signatures for the interventions, that may lead to targeted interventions. Despite these many positives, there were some limitations to this publication. First, the experiments were all completed in mice over a relatively short period of time—14 weeks of HFD-induced MetS PN. MetS in patients develops over years and is likely more complex than in the HFD mouse model; thus, these conclusions may not be applicable to patients with the disease. Second, there is no evidence that a ketogenic diet is a better intervention than just returning the animals to a standard diet. Third, there was no validation of the key findings from the RNAseq experiments. These were exploratory and will hopefully lead to future definitive studies, but conclusions from these studies are currently limited.

The publication by Eid et al. provides further evidence of the importance of MetS in the development of PN. This and other studies strongly suggest that MetS should be considered a prediabetes state and that DPN in T2DM is likely a result of not only elevated blood glucose but also features of MetS. The exact contribution of these different metabolic abnormalities remains unknown, but the similar clinical and pathologic features of PN in these conditions suggest that MetS abnormalities play an important role in DPN (16) (Fig. 1). The RNAseq experiments of Eid et al. can further contribute to this investigation by comparing the gene expression within peripheral nerves and muscle in this mouse model of MetS with that in models of DPN, with and

without lifestyle interventions. For validation, data from these studies should also be compared with data from patients with MetS and T2DM (17,18). Eid et al. also provide further evidence that lifestyle changes (i.e., dietary modifications or exercise) are highly effective at reversing MetS PN. Lifestyle modifications unfortunately are difficult to study in a placebo-controlled trial and are plagued by poor compliance over the long term. For this reason, discovering the critical downstream mechanisms from MetS PN and effective interventions are critical to devising medical treatments for this disabling condition. Future studies should be directed toward confirming the implicated genes and pathways from the RNAseq studies and discovering translatable drugs for treating this disease.

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